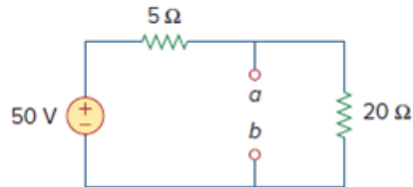


Problem

The Norton current at terminals a and b of the circuit in Fig. 4.67 is:

- (a) 10 A
- (b) 2.5 A
- (c) 2 A
- (d) 0 A

Figure. 4.67:



Step-by-step solution

Step 1 of 1

Refer to Figure 4.67 in the textbook.

Find the Norton current at terminals a and b of the circuit.

Short the terminals a and b to find the current through them.

The $20\ \Omega$ resistance in the circuit vanishes, since there is short circuit across it. Therefore, the short circuit current is,

$$\begin{aligned} i_{sc} &= \frac{V_{oc}}{R} \\ &= \frac{50\text{ V}}{5\ \Omega} \\ &= 10\text{ A} \end{aligned}$$

Thus, the value of Norton current through the terminals a and b is $\boxed{10\text{ A}}$.

Hence, the correct option is, $\boxed{(a)}$.